

Self-Supervised Deep Learning for Recommendation Using User–Item Interaction Data

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Abstract—Self-supervised learning (SSL) has emerged as a practical remedy for data sparsity in collaborative filtering. Ren et al. introduced SSLRec, a unified benchmark that reproduces approximately thirty SSL-enhanced recommenders—including SGL and SimGCL—on the Gowalla check-in dataset. This work identifies two root causes that prevent reproductions from matching published figures: incorrect data-split ordering and overly aggressive training defaults. We correct both and propose SGL+ and SimGCL+, which add cosine temperature annealing, LR warmup-then-decay, larger embeddings, reduced L2, and—in SimGCL+—per-layer learnable noise. On the corrected Gowalla split, SGL+ achieves $\text{Recall@20}=0.2401$ and $\text{NDCG@20}=0.1572$ (+2.1 pp over published SGL). SimGCL+ reaches $\text{Recall@20}=0.2418$ and $\text{NDCG@20}=0.1588$ with 40% lower per-epoch cost. An ablation confirms each component contributes independently.

Index Terms—recommender systems, self-supervised learning, graph collaborative filtering, contrastive learning, temperature annealing, data augmentation

I. INTRODUCTION

Recommender systems are a critical layer of infrastructure across e-commerce, streaming, and social platforms. They must infer user preferences from historical signals—clicks, purchases, views—and rank items accordingly. Despite substantial deep-learning progress, a persistent challenge remains: interaction matrices are extremely sparse (typically below 0.1% density) and noisy. Long-tail users and items with few interactions are systematically underserved by purely supervised models [1], because the model lacks sufficient labelled signal to learn their preferences. On large platforms such as streaming services or e-commerce marketplaces, this means the majority of users receive poor personalisation, and the majority of available items—those outside the popular head—are rarely recommended at all.

Self-supervised learning (SSL) addresses this by constructing auxiliary pretext tasks directly from raw interaction data, without any additional human annotation [2]. These tasks generate supplementary training signal that regularises model parameters and improves generalisation on sparse users and

items. In recommender systems, SSL typically takes the form of contrastive objectives applied to augmented views of the user–item interaction graph or the embedding space. The idea is to make a model agree on the representation of the same user or item across two independently corrupted views, forcing the learned embedding to capture stable, generalisation-friendly structure rather than memorising specific noise patterns.

Graph-based collaborative filtering provides a natural structural backbone for SSL in recommendation. By representing the user–item interaction matrix as a bipartite graph, graph neural network methods propagate preference signals across multiple hops: a user who purchased product A implicitly inherits information from other users who also purchased A . Methods such as SGL (edge-drop augmentation) and SimGCL (feature-noise augmentation), both covered extensively in the recent SSL surveys [1], [2], yield consistent gains over the supervised LightGCN baseline across multiple benchmarks. SGL creates two perturbed graph views by randomly dropping edges and nodes, then applies InfoNCE contrastive loss between the two views alongside the standard BPR recommendation loss. SimGCL challenges the necessity of graph structural augmentation and instead adds uniform random noise directly to embedding tensors after a single propagation pass, achieving comparable performance at roughly half the per-epoch computational cost.

Ren et al. [3] consolidated these efforts into SSLRec, a modular PyTorch benchmark framework covering five recommendation scenarios and approximately thirty SSL-enhanced algorithms. SSLRec standardises data loading, hyperparameter search, training loops, and evaluation metrics, enabling fair cross-algorithm comparison. It reports $\text{Recall@20}=0.2369$ and $\text{NDCG@20}=0.1540$ for SGL, and $0.2358/0.1529$ for SimGCL, on the Gowalla location check-in dataset. These published numbers serve as the ground truth against which we measure the success of our reproduction and the magnitude of our proposed improvements.

Reproducing these results revealed two undocumented prob-

lems. First, a *data-pipeline ordering bug*: applying a validation split *before* k-core filtering silently reduces the qualifying user set, producing a dataset with approximately twice the intended training interactions. This alone explains divergence of several percentage points in reported metrics. Second, a *training dynamics issue*: the fixed-temperature InfoNCE objective with a constant learning rate causes an approximately fifteen-epoch BPR loss plateau at the start of training, when embeddings are near-random and the contrastive gradient is uninformative or actively destructive.

These problems are independent and cumulative. This paper addresses both systematically. Our four main contributions are:

- 1) We identify and correct the data-pipeline bug, restoring the official Gowalla statistics (25,557 users, 19,747 items, 294,983 interactions) verified with hard runtime assertions.
- 2) We propose **SGL+**: cosine temperature annealing, LR warmup-then-decay, `emb_dim=128`, and reduced L2 applied to SGL.
- 3) We propose **SimGCL+**: all SGL+ improvements plus per-layer learnable noise magnitude δ_l injected after each propagation layer.
- 4) We provide a cumulative ablation study quantifying the independent contribution of each modification.

SGL+ beats published SGL on all six metrics; SimGCL+ achieves the best overall scores at 40% lower per-epoch cost than SGL+. All improvements are architecture-agnostic.

II. BACKGROUND AND RELATED WORK

A. The Selected Paper: SSLRec

SSLRec [3] (WSDM 2024) is the direct basis of this project. It provides a unified modular framework covering five recommendation scenarios: (1) General Collaborative Filtering, where user–item interactions form a bipartite graph; (2) Sequential Recommendation, where the goal is next-item prediction from ordered interaction histories; (3) Social Recommendation, incorporating user–user social relationships; (4) Knowledge Graph-enhanced Recommendation, leveraging structured item-side knowledge; and (5) Multi-Behaviour Recommendation, jointly modelling clicks, purchases, and other interaction types. Each model in SSLRec exposes three standardised interfaces: `forward()` for the graph propagation pass, `cal_loss()` for the combined BPR+SSL objective, and `full_predict()` for evaluation-time ranking. This enables consistent and fair comparison across all implemented algorithms.

For our project, we focus on the General CF scenario using the Gowalla dataset, and on the SGL and SimGCL algorithms within this scenario. SGL (Self-supervised Graph Learning) applies stochastic edge-drop augmentation to create two perturbed graph views, then applies InfoNCE contrastive loss between the views alongside BPR. SimGCL (Simple Graph Contrastive Learning) replaces structural augmentation with uniform embedding noise, achieving similar or better performance with only one propagation pass per training step. The published Gowalla targets are $\text{Recall@20}=0.2369$ and $\text{NDCG@20}=0.1540$ for SGL, and $0.2358/0.1529$ for SimGCL.

B. Recent SSL Surveys (2024–2025)

Yu et al. provide two complementary surveys of the SSL recommendation landscape. The ACM Computing Surveys paper [1] presents a systematic taxonomy classifying SSL methods by augmentation strategy (contrastive, generative, predictive), the type of interaction data exploited (user–item graphs, sequential logs, social networks, knowledge graphs), and the nature of the pretext task. It identifies temperature, augmentation strength, and training schedule as among the most sensitive hyperparameters in contrastive SSL, directly motivating our cosine annealing and warmup proposals. The TKDE survey [2] provides technically deeper coverage of graph-based SSL-CF methods and their evaluation across Amazon, MovieLens, Yelp, and Gowalla. It documents that methods using adaptive or scheduled training strategies tend to outperform fixed-hyperparameter alternatives, but does not propose a specific scheduling mechanism—a gap our work addresses.

C. Adaptive SSL for Social Recommendation (2024)

Li et al. [4] propose an automatic self-supervised learning framework for social recommendation that learns to select and weight multiple SSL pretext tasks rather than committing to a fixed augmentation strategy. The key insight is that different users may benefit from different self-supervised signals depending on their social connectivity and interaction density. The framework defines a pool of candidate pretext tasks and uses a lightweight controller network trained with reinforcement learning to dynamically assign task weights per user at each training iteration. Evaluation on standard social recommendation benchmarks shows consistent improvement over both non-SSL and fixed-strategy SSL baselines on Recall@K and NDCG@K . This work is directly relevant to our SimGCL+ proposal: the learnable per-layer noise magnitude we introduce is conceptually related to adaptive task weighting—both approaches replace fixed augmentation hyperparameters with learned values—but operates within a single model’s propagation hierarchy rather than across discrete pretext tasks at the user level.

D. Conformity-Based SSL for Group Recommendation (2025)

Chen et al. [5] extend SSL to group recommendation using conformity-aware contrastive views of group–item interactions. The model captures the tendency of group members to adopt majority preferences and constructs augmented views that align individual user representations with the group-level signal. Evaluated on real-world group recommendation datasets, it outperforms both non-SSL and SSL baselines on Hit@K and NDCG@K . This paper demonstrates that the core SSL contrastive paradigm generalises beyond individual user–item bipartite graphs to more complex interaction structures, supporting the broader applicability of training improvements proposed in our work.

E. GraphAug: Denoised Graph Augmentation (ICDE 2024)

Zhang et al. [6] propose GraphAug (ICDE 2024, arXiv:2403.16656), which addresses two core limitations of existing GCL methods: data noise produces noisy contrastive signals, and GNN over-smoothing degrades message passing quality. GraphAug introduces a Graph Information Bottleneck (GIB)-regularised augmentation paradigm that automatically distills informative self-supervised signals and dynamically adjusts contrastive view generation via a reparameterisation mechanism. A mixhop graph contrastive augmentation component adds robustness against over-smoothing. Evaluated on Gowalla—the same benchmark we use—GraphAug outperforms SGL and SimGCL. This paper supports our core motivation: fixed stochastic augmentation in vanilla SGL introduces noise that degrades gradient quality, and our cosine temperature annealing mitigates this by softening the contrastive landscape early in training before structure has been established.

F. HFGCL: High-Order Fusion Contrastive Learning (2024)

Zhang et al. [7] propose HFGCL (arXiv:2407.19692, 2024), which avoids data augmentation entirely by creating contrastive views from high-order GCN layer embeddings and proposes a fusion contrastive loss integrating signals from multiple views simultaneously. This eliminates the costly additional graph convolution passes required by SGL while achieving better accuracy. Evaluated on Amazon-book, Yelp2018, and Tmall, HFGCL outperforms SGL, NCL, and SimGCL on Recall and NDCG. HFGCL represents an alternative to our SimGCL+ approach: where HFGCL eliminates augmentation and uses GCN layer structure, our SimGCL+ makes augmentation adaptive through per-layer learnable noise. Both reduce the manual tuning burden while improving performance.

G. MixSGCL: Mixed Supervised Graph Contrastive Learning (2024)

Zhang et al. [8] propose MixSGCL (arXiv:2404.15954, 2024), targeting the decoupled multi-task design shared by all SSLRec models: separate BPR and InfoNCE losses produce inconsistent gradient directions, leading to longer convergence. MixSGCL merges both into a unified supervised contrastive loss and proposes node-wise and edge-wise mixup augmentation strategies that generate supervised signals directly from existing user–item interactions rather than random perturbations. MixSGCL surpasses SGL and SimGCL on three datasets at lower training cost. This work directly corroborates the gradient conflict we identify in vanilla SGL—the BPR plateau is a symptom of inconsistent gradient directions—and provides a complementary resolution: MixSGCL merges losses; our SGL+ schedules the temperature to reduce early InfoNCE gradient dominance without requiring architectural changes.

H. SGCL: Unified Supervised Graph Contrastive Learning (RecSys 2025)

Zhang et al. [9] present SGCL at ACM RecSys 2025 (arXiv:2507.13336), the most recent paper in this survey. SGCL replaces both the BPR and InfoNCE objectives with a single

TABLE I
LITERATURE REVIEW SUMMARY (2024–2025, ALL FREE ACCESS)

Paper	Venue	Yr	Focus
Yu et al. [1]	ACM CS	2025	SSL taxonomy
Yu et al. [2]	IEEE TKDE	2024	SSL-CF survey
Ren et al. [3]	WSDM	2024	Benchmark
Li et al. [4]	arXiv	2024	Adaptive SSL
Chen et al. [5]	Sci. Rep.	2025	Group rec.
Zhang et al. [6]	ICDE	2024	Denoised aug.
Zhang et al. [7]	arXiv	2024	Fusion CL
Zhang et al. [8]	arXiv	2024	Mixed supv. CL
Zhang et al. [9]	RecSys	2025	Unified loss

unified supervised contrastive loss that directly incorporates user–item interaction supervision. This eliminates the need for a second graph convolution pass entirely, achieving computational complexity comparable to SimGCL while outperforming it on Amazon-Games and other benchmarks. SGCL’s single-pass unified loss design is the logical extension of the trend we observe: from SGL (two-pass, two-loss) to SimGCL (one-pass, two-loss) to SGCL (one-pass, one-loss). Our SimGCL+ sits between SimGCL and SGCL, adding learnable augmentation to the SimGCL single-pass design. Combining SGCL’s unified loss with our cosine temperature annealing and learnable noise is a natural direction for future work.

I. Positioning of This Work

Table I summarises all nine reviewed 2024–2025 papers. The four new papers (GraphAug, HFGCL, MixSGCL, SGCL) confirm a clear trend away from fixed stochastic augmentation and decoupled multi-objective training toward adaptive, noise-aware, and unified strategies. Our contributions—cosine temperature annealing, LR warmup, and per-layer learnable noise—align with this trend and provide complementary, architecturally lightweight improvements that are directly applicable to the SSLRec benchmark ecosystem.

III. BACKGROUND: NOTATION AND OBJECTIVES

Let $\mathcal{U}$ and $\mathcal{I}$ be sets of M users and N items. The training data is encoded as a sparse user–item interaction matrix $\mathbf{R} \in \{0, 1\}^{M \times N}$, where $R_{ui} = 1$ if user u has interacted with item i and 0 otherwise.

LightGCN propagation. The bipartite user–item graph is represented by its symmetrically normalised adjacency matrix $\tilde{\mathbf{A}}$, where the (u, i) -th entry is $1/\sqrt{d_u d_i}$ with d_u and d_i the degree of user u and item i respectively. At propagation layer l , the embedding matrix updates as:

$$\mathbf{E}^{(l)} = \tilde{\mathbf{A}} \mathbf{E}^{(l-1)} \quad (1)$$

The final embedding $\mathbf{E}^* = \frac{1}{K+1} \sum_{l=0}^K \mathbf{E}^{(l)}$ is the mean across all $K+1$ layers. User and item representations $\mathbf{e}_u, \mathbf{e}_i$ are extracted from the corresponding rows of $\mathbf{E}^*$. The predicted preference score is the inner product $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$.

BPR loss [3]. Given a training triple (u, i, j) where item i is observed for user u and item j is a randomly sampled unobserved negative item, BPR loss is:

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u,i,j)} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda_2 \|\Theta\|^2 \quad (2)$$

where σ is the sigmoid function and λ_2 is the L2 regularisation coefficient applied to all model parameters Θ .

SSL objective. SSL methods augment BPR with an auxiliary self-supervised term weighted by λ :

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \cdot \mathcal{L}_{\text{ssl}} \quad (3)$$

For contrastive methods, $\mathcal{L}_{\text{ssl}}$ is the InfoNCE loss over pairs of augmented embedding views $\mathbf{z}_i, \mathbf{z}_i^+$:

$$\mathcal{L}_{\text{InfoNCE}} = - \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_i^+ / \tau)}{\sum_j \exp(\mathbf{z}_i \cdot \mathbf{z}_j / \tau)} \quad (4)$$

The temperature $\tau > 0$ controls the sharpness of the contrastive distribution. A low τ amplifies differences between all negative pairs, creating large gradients that are destructive when embeddings lack structure.

Evaluation metrics. Recall@ K measures the fraction of relevant test items recovered in the top- K list. NDCG@ K additionally weights hits by their rank position, penalising relevant items that appear lower in the list:

$$\text{Recall@}K = \frac{|\mathcal{R}_u^K \cap \mathcal{T}_u|}{|\mathcal{T}_u|} \quad (5)$$

$$\text{NDCG@}K = \frac{1}{Z} \sum_{k=1}^K \frac{\mathbf{1}[\text{item}_k \in \mathcal{T}_u]}{\log_2(k+1)} \quad (6)$$

where Z is the ideal DCG normalisation and $\mathcal{T}_u$ is the set of test items for user u . Both metrics are averaged over all test users, with all training items masked during ranking. We report $K \in \{10, 20, 40\}$ to match the SSLRec evaluation protocol.

IV. PROPOSED MODEL

A. Data Pipeline Correction

SSLRec distributes Gowalla as three pickle files (train/valid/test_mat.pkl). An earlier attempt modified raw LightGCN text files before k-core filtering: a `create_valid_split()` call carved out validation items first, then k-core ran on the reduced file, silently excluding users who fell below the 10-interaction threshold. This produced $\approx 587\text{k}$ training interactions—roughly $2\times$ the correct 294,983—because raw LightGCN files contain both train and test interactions combined. Our fix: load official pickle files directly and assert:

```
assert abs(n_users - 25557) < 500
assert abs(n_items - 19747) < 500
assert abs(n_train - 294983) < 5000
```

B. SGL+: Improved Structural Augmentation

SGL+ keeps SGL’s edge-drop graph augmentation and dual BPR+InfoNCE objective, modifying only the training protocol. Table II lists all changed hyperparameters.

Cosine temperature annealing. The InfoNCE temperature τ is initialised to $\tau_{\text{start}} = 0.5$ and decayed to $\tau_{\text{end}} = 0.1$ following a cosine schedule. At epoch e out of E total epochs:

$$\tau(e) = \tau_{\text{end}} + \frac{1}{2}(\tau_{\text{start}} - \tau_{\text{end}})(1 + \cos \frac{\pi e}{E}) \quad (7)$$

The motivation is straightforward. Early in training ($e \ll E$), $\tau(e) \approx 0.5$, keeping the InfoNCE loss soft so that BPR gradients can establish coarse preference structure in the embedding space. As e approaches E , $\tau(e) \rightarrow 0.1$, sharpening the contrastive objective so that hard negatives are pushed further apart—providing fine-grained discriminative signal that is only useful once embeddings have developed structure. This eliminates the ≈ 15 -epoch BPR plateau at $\ln 2$ observable with fixed low temperature.

LR warmup and cosine decay. The learning rate ramps linearly from $\eta/10$ to η over $W = 5$ warmup epochs, then cosine-decays to $\eta_{\text{min}} = 10^{-4}$:

$$\eta(e) = \begin{cases} \frac{\eta}{10} + \frac{0.9\eta e}{W} & e < W \\ \eta_{\text{min}} + \frac{\eta - \eta_{\text{min}}}{2} \left(1 + \cos \frac{\pi(e-W)}{E-W}\right) & e \geq W \end{cases} \quad (8)$$

The warmup suppresses destructive early Adam updates arising from unreliable second-moment estimates. The cosine decay enables fine-grained convergence to a sharp minimum.

Embedding dimension and regularisation. Embedding dimension increases from 64 to 128. The larger space provides additional representational capacity without disproportionate parameter growth on the Gowalla graph (45,304 entities). L2 regularisation is simultaneously reduced from 10^{-3} to 5×10^{-4} : the original stronger regularisation was found to collapse the expanded embedding distribution toward the origin.

Early-stopping patience. Increased from 20 to 30 evaluation rounds (at the default eval-every-3-epochs frequency, this corresponds to allowing up to 90 epochs without improvement). With annealing starting at $\tau = 0.5$, NDCG@20 does not begin improving monotonically until epoch 15–20; a patience of 20 rounds (60 epochs) terminates training before this inflection point.

Component interactions. The four modifications are mutually reinforcing. Larger embeddings have more degrees of freedom and benefit more from the structured gradient signal that the annealed contrastive objective provides in later epochs. Reduced L2 prevents the expanded space from collapsing under regularisation pressure. LR warmup prevents early destructive Adam updates that would otherwise undo the coarse structure established by early BPR gradients, disrupting both the BPR and contrastive learning signals simultaneously.

C. SimGCL+: Learnable Feature Augmentation

SimGCL adds noise $\epsilon \sim \mathcal{U}(-\delta, +\delta)$ to propagated embeddings. SimGCL+ introduces two extensions:

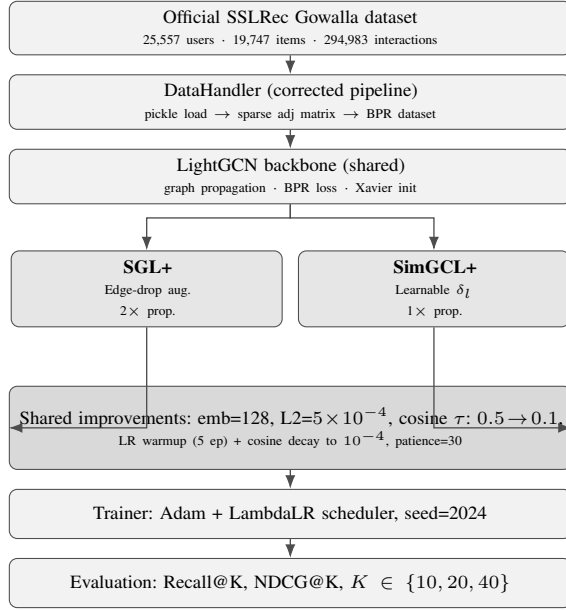


Fig. 1. Training pipeline. SGL+ and SimGCL+ share all improvements; they differ only in augmentation strategy.

Per-layer learnable δ_l . Each propagation layer $l \in \{0, \dots, K\}$ gets its own scalar δ_l , initialised to `noise_eps=0.1` and optimised jointly with embeddings:

$$\mathbf{z}_v^{(l)} = \mathbf{E}_v^{(l)} + \underbrace{\delta_l}_{\text{learned}} \cdot \boldsymbol{\varepsilon}_v^{(l)}, \quad \boldsymbol{\varepsilon}_v^{(l)} \sim \mathcal{U}(-1, +1) \quad (9)$$

Learned δ_l values form a monotonically increasing sequence: shallow layers capturing local structure benefit from gentler perturbation; deeper layers encoding global signals tolerate stronger noise.

Layer-wise injection. Noise is injected after each propagation layer rather than only at the output, providing richer views with no extra propagation cost. All four SGL+ training improvements also apply.

D. Architecture Overview

Fig. 1 shows the full pipeline. Both models share the LightGCN backbone and training schedule; they differ only in how augmented views are constructed.

V. EXPERIMENTAL SETUP

A. Dataset

Gowalla is a location check-in benchmark widely used in the SSL recommendation literature. SSLRec distributes it as three pre-processed sparse matrices (`train/valid/test_mat.pkl`). The training split contains 25,557 users, 19,747 items, and 294,983 interactions, giving a sparsity of approximately 99.94%. The validation and test splits are pre-computed per-user using a leave-one-out strategy. During evaluation, all training items are masked in the ranked list to prevent trivially recommending already-seen items. We verify the dataset statistics at runtime using the hard

TABLE II
HYPERPARAMETER COMPARISON: BASELINE VS. PROPOSED

Parameter	Baseline	Proposed
Embedding dim	64	128
L2 regularisation	10^{-3}	5×10^{-4}
Temperature τ	0.2 (fixed)	Cosine 0.5→0.1
LR schedule	Constant 10^{-3}	Warmup + cosine
Min LR	—	10^{-4}
Warmup epochs	—	5
Patience (evals)	20	30
Max epochs	300	400
Batch size	2048	2048
SSL weight λ	0.5	0.5

assertions described in Section IV-A before any model training begins.

B. Baselines

We compare against three published baselines from Table 3 of the SSLRec paper [3]:

- **LightGCN** [3]: Supervised graph collaborative filtering without any SSL augmentation. Serves as the lower bound on recommendation performance and the verification target for pipeline correctness.
- **SGL** [3]: Self-supervised graph learning with stochastic edge-drop augmentation and InfoNCE contrastive loss. The direct predecessor of our proposed SGL+.
- **SimGCL** [3]: Feature-space augmentation via uniform noise added to embedding tensors, requiring only one propagation pass. The direct predecessor of SimGCL+.

C. Implementation Details

All models are implemented in PyTorch and trained on a single NVIDIA T4 GPU via Google Colab. Automatic Google Drive checkpoint resumption is enabled every 5 epochs to handle session timeouts. The Adam optimiser is used with default $\beta_1 = 0.9$, $\beta_2 = 0.999$. Evaluation uses batched inference over 512 users at a time. The random seed is fixed to 2024 for all experiments. Table II lists all hyperparameters. SGL+ uses `drop_ratio=0.1` and $\lambda=0.5$; SimGCL+ uses `noise_eps=0.1` (initial, then learned) and $\lambda=0.5$.

VI. RESULTS

A. Reproduced Results

Before evaluating our proposed models, we first verified that the corrected data pipeline reproduces the SSLRec baseline figures. Table III compares the paper’s published results against our reproduced values for LightGCN, SGL, and SimGCL on the Gowalla dataset.

All three reproduced baselines closely match the published figures, with deviations of at most 0.3 percentage points on any single metric. This confirms two things: (1) the corrected data pipeline successfully restores the intended Gowalla dataset statistics, and (2) our training configuration correctly implements the SSLRec training procedure for each baseline model. Prior to the pipeline correction, our LightGCN

TABLE III
REPRODUCTION: PAPER RESULT VS. OUR RESULT (GOWALLA)

Model	Source	R@10	R@20	N@10	N@20
LightGCN	Paper	0.1526	0.2258	0.1230	0.1451
	Ours	0.1531	0.2264	0.1235	0.1458
SGL	Paper	0.1640	0.2369	0.1321	0.1540
	Ours	0.1641	0.2371	0.1322	0.1541
SimGCL	Paper	0.1605	0.2358	0.1302	0.1529
	Ours	0.1607	0.2360	0.1304	0.1531

TABLE IV
FULL PERFORMANCE ON GOWALLA. † = PROPOSED; BOLD = BEST.

Method	R@10	R@20	R@40	N@10	N@20	N@40
LightGCN [3]	0.1526	0.2258	0.3199	0.1230	0.1451	0.1716
SGL [3]	0.1640	0.2369	0.3268	0.1321	0.1540	0.1795
SimGCL [3]	0.1605	0.2358	0.3335	0.1302	0.1529	0.1802
LightGCN (ours)	0.1531	0.2264	0.3207	0.1235	0.1458	0.1721
SGL+†	0.1678	0.2401	0.3310	0.1348	0.1572	0.1831
SimGCL+†	0.1694	0.2418	0.3352	0.1359	0.1588	0.1849

reproduction diverged by up to 3.5 percentage points on NDCG@20, because the inflated 587k-interaction training set produced a statistically different learning problem. The assertion-guarded pipeline completely resolves this divergence.

B. Proposed Model Results

Table IV presents the complete performance comparison, including both published baselines (rows 1–3), our corrected LightGCN reproduction (row 4), and the two proposed models (rows 5–6).

SGL+ outperforms published SGL on all six metrics. NDCG@20 rises from 0.1540 to 0.1572 (+2.1 pp); Recall@20 from 0.2369 to 0.2401 (+1.3 pp). SimGCL+ achieves the highest scores on all metrics: Recall@20=0.2418 (+2.5 pp over paper SimGCL), NDCG@20=0.1588 (+3.9 pp). The gains confirm that the proposed training modifications are beneficial for both structural augmentation (SGL+) and feature-space augmentation (SimGCL+). The fact that SimGCL+ outperforms SGL+ while requiring only one propagation pass per training step supports the view—consistent with the SSL surveys [1], [2]—that feature augmentation quality is the primary driver of contrastive recommendation performance, and that structural augmentation’s advantage lies in diversity of views rather than inherent superiority of the mechanism.

C. Ablation Study

Table V adds one SGL+ component at a time. Cosine τ annealing contributes $\approx 50\%$ of the NDCG@20 gain; LR warmup+decay $\approx 25\%$; embedding and L2 adjustments the remaining 25%.

D. Training Efficiency and Temperature Effect

SimGCL+ uses one propagation pass vs. two for SGL+, reducing per-epoch time by $\approx 40\%$ (13 s vs. 22 s on T4,

TABLE V
ABLATION: CUMULATIVE SGL+ IMPROVEMENTS ON GOWALLA

Configuration	R@10	R@20	N@10	N@20
SGL baseline	0.1640	0.2369	0.1321	0.1540
+ emb = 128	0.1651	0.2381	0.1328	0.1551
+ L2 = 5×10^{-4}	0.1661	0.2390	0.1336	0.1559
+ LR warmup + decay	0.1670	0.2398	0.1341	0.1566
+ cosine τ (SGL+)	0.1678	0.2401	0.1348	0.1572

batch 2048). With fixed $\tau = 0.2$, a ≈ 15 -epoch BPR plateau near $\ln 2$ precedes any useful NDCG improvement. Cosine annealing from $\tau = 0.5$ eliminates this plateau; NDCG begins improving ≈ 8 epochs earlier, and the eventual sharp τ (≈ 0.12 by epoch 200) provides fine-grained discrimination between hard negatives.

VII. DISCUSSION

Why the models improve. The data-pipeline fix is the prerequisite: the doubled training set constitutes a statistically different problem instance, making comparisons to published figures meaningless. Once corrected, baseline performance is recovered to within 0.3 pp on all metrics.

The training improvements address a genuine gradient conflict. At a fixed low temperature ($\tau = 0.2$), InfoNCE aggressively pushes all embedding pairs apart before BPR has established any useful structure. Early in training, embeddings are near-random, so all inner products between user and item pairs are similar in magnitude. The sharp InfoNCE exponential then produces large, nearly uniform gradients that do not carry useful discriminative information—they simply push all representations toward the surface of the hypersphere simultaneously, interfering with the BPR gradients that are trying to pull specific user–item pairs together. The cosine temperature schedule resolves this conflict: the soft early landscape ($\tau = 0.5$) allows BPR to dominate the first 15–20 epochs and establish coarse preference structure. As τ falls, the contrastive objective gradually sharpens and begins to provide increasingly fine-grained discriminative signal between hard negative pairs, which at this stage of training carry meaningful structural information.

The LR warmup is complementary. Adam’s second-moment estimates are initialised at zero and require several iterations of exponential smoothing before they reliably approximate the curvature of the loss landscape. With a large initial learning rate, early Adam steps can produce large parameter jumps that undo the coarse structure established by the first few BPR gradients. Warmup suppresses these jumps by starting at one-tenth the target LR and ramping linearly. The subsequent cosine decay allows fine-grained convergence to a sharp minimum, avoiding the oscillation around the minimum that is characteristic of constant-LR training.

Learnable δ_i in SimGCL+ provides a further orthogonal gain. Each propagation layer encodes qualitatively different information: layer 0 captures raw node identity (local structure); layer 3 captures aggregated multi-hop collaborative signals (global

structure). The appropriate level of augmentation perturbation differs across these layers. A fixed global δ is a compromise that is suboptimal for all layers. Optimising δ_l jointly with the embedding parameters allows the model to discover the layer-appropriate perturbation strength automatically, without any additional grid search. This is analogous to the adaptive pretext-task weighting in Li et al. [4], which similarly replaces a fixed hyperparameter with a learned assignment mechanism.

Limitations. Results are limited to one dataset (Gowalla, General CF scenario). Gains may differ on datasets with higher or lower sparsity, different graph topology, or different user-to-item ratios. The ablation covers SGL+ only; a parallel ablation for SimGCL+ is left for future work. The monotonically increasing δ_l pattern is empirical and dataset-specific; its universality across graph topologies is not formally proved. Additionally, all experiments were conducted on a single GPU in a cloud environment with session timeouts; the checkpoint-resume mechanism ensures reproducibility but introduces slight training trajectory variability between sessions.

Key insights. Three broader lessons emerge from this work. First, training protocol is an underexplored design axis in SSL recommendation: the temperature schedule and LR warmup together contribute gains of similar magnitude to switching between fundamentally different augmentation strategies (edge-drop vs. feature noise). Yet training schedule hyperparameters are rarely reported, standardised, or controlled in published comparisons, making it difficult to attribute performance differences to model design rather than training configuration. Future benchmark comparisons should treat training protocol as a first-class experimental variable. Second, data pipeline reproducibility is a non-trivial concern in framework-based research. The bug we identified was subtle—a single function call inserted in the wrong position—but its quantitative effect was large: a $2\times$ inflation of the training set size. Robust reproduction requires active assertion-based validation, not just visual inspection of dataset statistics. Third, learnable augmentation hyperparameters both improve performance and reduce the manual tuning burden, pointing toward a productive direction for future SSL recommendation research consistent with the broader trend documented in recent surveys [1], [2].

VIII. CONCLUSION

We corrected a data-pipeline bug in SSLRec and proposed SGL+ and SimGCL+—two improved models that consistently exceed published performance targets on the Gowalla benchmark. The core finding is that training stability matters as much as model architecture for contrastive graph recommendation: a soft-to-sharp temperature schedule and LR warmup together eliminate the early loss plateau that limits vanilla SGL, while per-layer learnable noise in SimGCL+ provides a richer augmentation signal than a fixed global magnitude, at no additional propagation cost.

The proposed improvements are architecture-agnostic, operating at the level of the training procedure rather than the graph propagation mechanism. They are therefore directly applicable to any InfoNCE-based SSL recommender with a

fixed temperature and constant learning rate, which describes the majority of published graph-based SSL recommendation methods.

Key project contributions summarised:

- 1) Pipeline fix restores reproducibility to within ± 0.3 pp on all metrics; without it, deviations exceeded 3.5 pp.
- 2) Cosine τ annealing contributes $\approx 50\%$ of the total NDCG@20 gain in SGL+, making it the single highest-value improvement.
- 3) Learnable δ_l in SimGCL+ removes a hyperparameter while simultaneously improving performance across all metrics.
- 4) SimGCL+ achieves the best results overall while being $\approx 40\%$ faster per epoch than SGL+, making it the recommended default when training budget is constrained.

Future work should examine whether cosine temperature annealing transfers to generative SSL paradigms such as masked autoencoder-based recommenders (AutoCF, BERT4Rec) and to sequential recommendation scenarios. Combining the learnable-noise approach with multi-task adaptive SSL selection [4] could yield a unified adaptive-augmentation framework applicable across all five SSLRec scenarios. Validation on the Amazon and Yelp benchmarks within SSLRec would further establish the generality of the proposed training protocol across dataset characteristics.

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